

EDUCATION

- **University of California, Los Angeles**

Ph.D. Student in Computer Science (incoming)
M.S. in Computer Science
B.S. in Computer Science

Los Angeles, CA

Begins Sept. 2026
Sept. 2024 – June 2026
Oct. 2020 – Mar. 2024

◦ Coursework: Advanced Deep Learning, Natural Language Processing, Neurosymbolic Reasoning, Human-Computer Interaction

EXPERIENCE

- **Computation and Discovery Lab, UCLA**

Graduate Student Researcher

Los Angeles, CA

Feb. 2025 – Present

◦ Designed Syrup, a concept-role-graph intermediate representation for bidirectional human-AI scientific reasoning; co-first-authored the resulting paper

◦ Built the accompanying interactive system and evaluated it through qualitative and quantitative user studies with researchers and PhD students
- **Excavaite Inc**

CTO & Co-founder

Palo Alto, CA

Feb. 2024 – May 2026

◦ Built a terabyte-scale patent corpus with RaBitQ-indexed vector search on Postgres and VectorChord, powering retrieval for agentic workflows

◦ Designed agentic systems for patent leakage prevention, patent drafting, and claim chart generation; built automated evals to gate model and prompt changes

◦ Secured a six-figure Fortune 500 pilot and raised pre-seed funding

◦ Hired and managed an 8-person engineering team, owning architecture and technical direction end to end
- **VAIL Lab, UCLA**

Undergraduate Researcher

Los Angeles, CA

June 2023 – June 2024

◦ Researched diffusion models for synthetic driving scene generation, resulting in SimGen (NeurIPS 2024)

◦ Created a unified terabyte-scale driving dataset by combining 4K YouTube footage with open-source datasets
- **Confluent**

Software Engineer Intern, Kafka Metadata Team

Mountain View, CA

June 2023 – Aug. 2023

◦ Built a last-resort Java metadata recovery tool, cutting Kafka cluster recovery time from hours to minutes
- **Meta**

Software Engineer Intern, Metrics Platform Governance

Menlo Park, CA

June 2022 – Aug. 2022

◦ Created a centralized Hack (PHP) code-generation framework, cutting data onboarding time from ~1 hr to ~5 min; parallelized generation for a further 4x speedup
- **Agrofocal Technologies**

Machine Learning Intern (part-time)

San Jose, CA

June 2021 – Nov. 2023

◦ Implemented a mixed-precision quantization pipeline, increasing inference throughput by over 80% and halving latency

◦ Implemented a state-of-the-art PyTorch architecture, improving object detection accuracy by over 0.15 mAP

◦ Removed the need for a discrete GPU on edge devices via a hardware-acceleration SDK, saving \$1,000s per device

SELECTED PUBLICATIONS

- **Towards Bidirectional Human–AI Scientific Reasoning: From Concept-Role Graphs to Negotiated Interpretation.** *M.S. thesis, UCLA, June 2026.*

• **SimGen: Simulator-conditioned Driving Scene Generation.** *NeurIPS 2024* (second author).